

[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #7

#LP: There is a circular park of diameter 65 m as shown in the following figure, where AB is a diameter. An entry gate is to be constructed at a point P on the boundary of the park such that distance of P from A is 35 m more than the distance of P from B. Find distance of point P from A and B respectively. (CBSE 2025)

ATQ $PA = 35 + PB$

Let $PB = (x) \text{ m}$

$PA = (35 + x) \text{ m}$

Since angle in semi-circle = 90°

$\angle APB = 90^\circ$

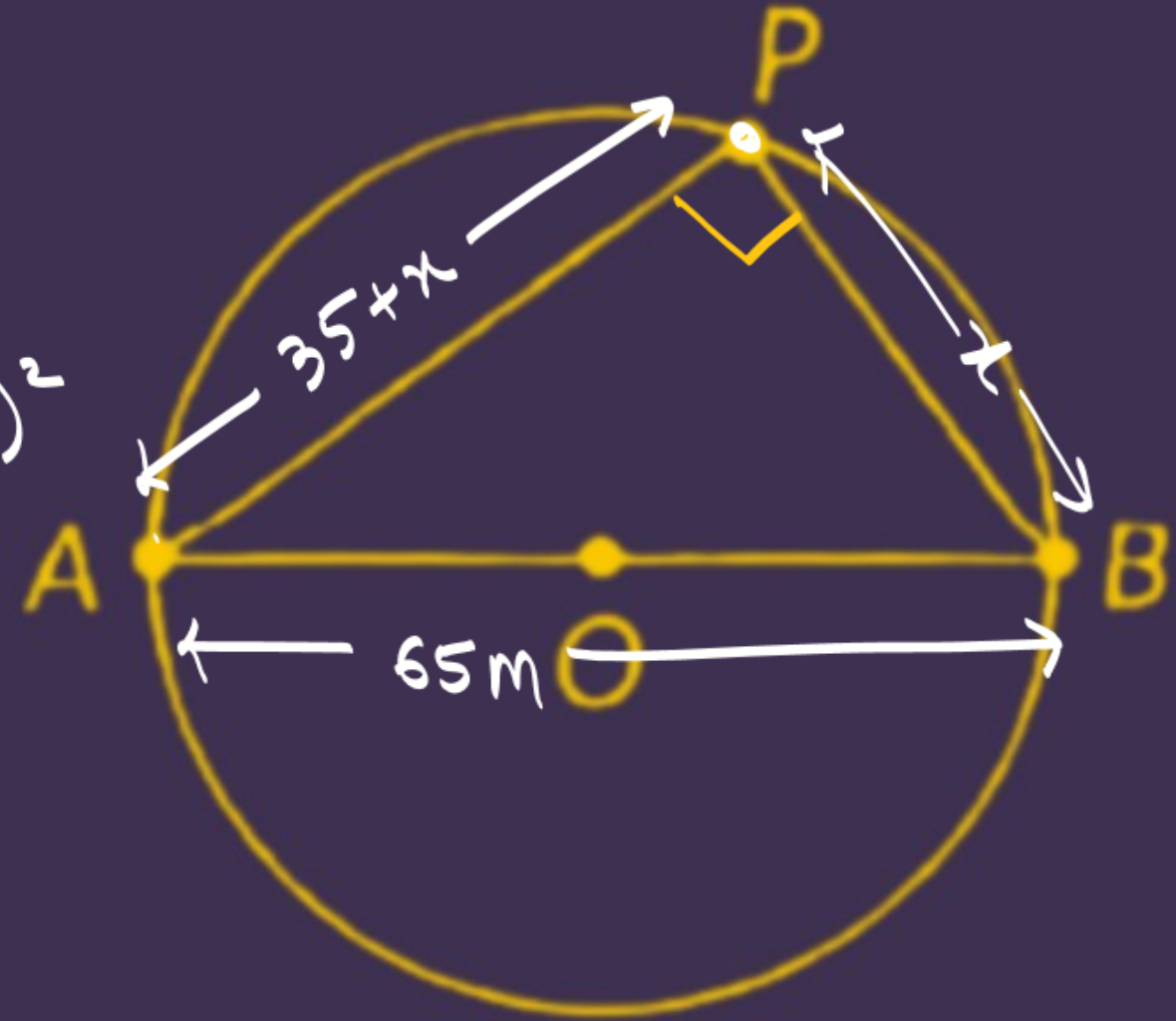
This is a right Δ .

Pyth

$$H^2 = P^2 + B^2$$

$$(65)^2 = (x)^2 + (x + 35)^2$$

Solve



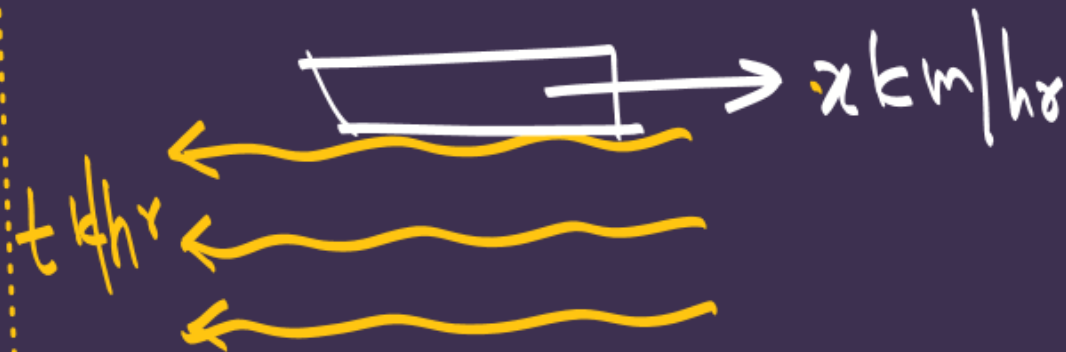
STILL WATER



speed of stream $(t) \Rightarrow 0$

speed of boat in still water $\Rightarrow x \text{ km/hr}$

UPSTREAM



speed of stream $\Rightarrow t \text{ km/hr}$

speed of boat in still water $\Rightarrow x \text{ km/hr}$

Net speed of boat $\Rightarrow (x - t) \text{ km/hr}$

Upstream speed

DOWNSTREAM



Net speed of boat $\Rightarrow (x + t) \text{ km/hr}$

Downstream speed

Stream Wale Problem :

#LP: The speed of a boat in still water is 8km/hr. It can go 15 km upstream and 22 km downstream in 5 hours. Find the speed of the stream.

$$d = s \times t$$
$$\boxed{\frac{d}{s} = t}$$

$x = 8 \text{ km/hr}$
 $t = ?$
Upstream

$$d_u = 15 \text{ km}$$
$$S_u = (8 - t) \text{ km/hr}$$

$$t_u = \frac{d_u}{S_u}$$

$$\boxed{t_u = \frac{15}{8 - t}}$$

Downstream

$$d_d = 22 \text{ km}$$
$$S_d = (8 + t) \text{ km/hr}$$

$$t_d = \frac{d_d}{S_d}$$

$$\boxed{t_d = \left(\frac{22}{8 + t} \right)}$$

Total time = 5 hr

$$t_u + t_d = 5$$

$$\boxed{\frac{15}{8 - t} + \frac{22}{8 + t} = 5}$$

LCM solve
 $t = 2$ ✓

#LP: A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream. (CBSE 2020)

$$x = 18 \text{ km/h}$$

Let speed of stream $\Rightarrow t$ km/h
??

ATQ

$$t_u = 1 + t_d$$

$$\Rightarrow \frac{d_u}{s_u} = 1 + \frac{d_d}{s_d}$$

$$\Rightarrow \frac{24}{18-t} - \frac{24}{18+t} = 1$$

$$\Rightarrow 24 \left(\frac{1}{18-t} - \frac{1}{18+t} \right) = 1$$

L.C.M

$$\begin{aligned} d &= s \times t \\ \frac{d}{s} &= t \end{aligned}$$

Age Wale Problem :

#LP: Ashu is x years old, while his mother, Mrs. Veena, is x^2 years old. Five years hence Mrs. Veena will be three times old as Ashu. Find their present ages.

	Present	5 yrs later
Ashu	x	$\rightarrow (x+5)$
Veena	x^2	$\rightarrow (x^2+5)$

ATQ $\Rightarrow (x^2+5) = 3(x+5)$

$$x^2 + 5 = 3x + 15$$

$$x^2 - 3x - 10 = 0$$

$$x = 5$$

← Ashu

$$x^2 = 25$$

← Veena Present

#LP: The sum of the ages of a man and his son is 45 years. Five years ago, the product of their ages was four times the man's age at the time. Find their present ages.

	5 yrs ago	Present
Son	$\frac{(x-5)}{45-x-5}$	x
Father	$\frac{(40-x)}{(45-x)}$	$(45-x)$ (ATQ)

ATQ

$$(x-5)(40-x) = 4(45-x)$$

↓
Q. Verd

#LP: The product of Ramu's age (in years) five years ago and his age (in years) nine years later is 15. Determine Ramu's present age.

5 yrs ago	Present	9 yrs later
$(x-5)$	x (let)	$(x+9)$

ATO $(x-5)(x+9) = 15$

↓
Solve x ✓

Length Wale Problem :

#LP: The diagonal of a rectangular field is 60 meters more than the shorter side. If the longer side is 30 meters more than the shorter side, find the sides of the field.

$$AC = 60 + AB$$

Let $AB = (x)m$

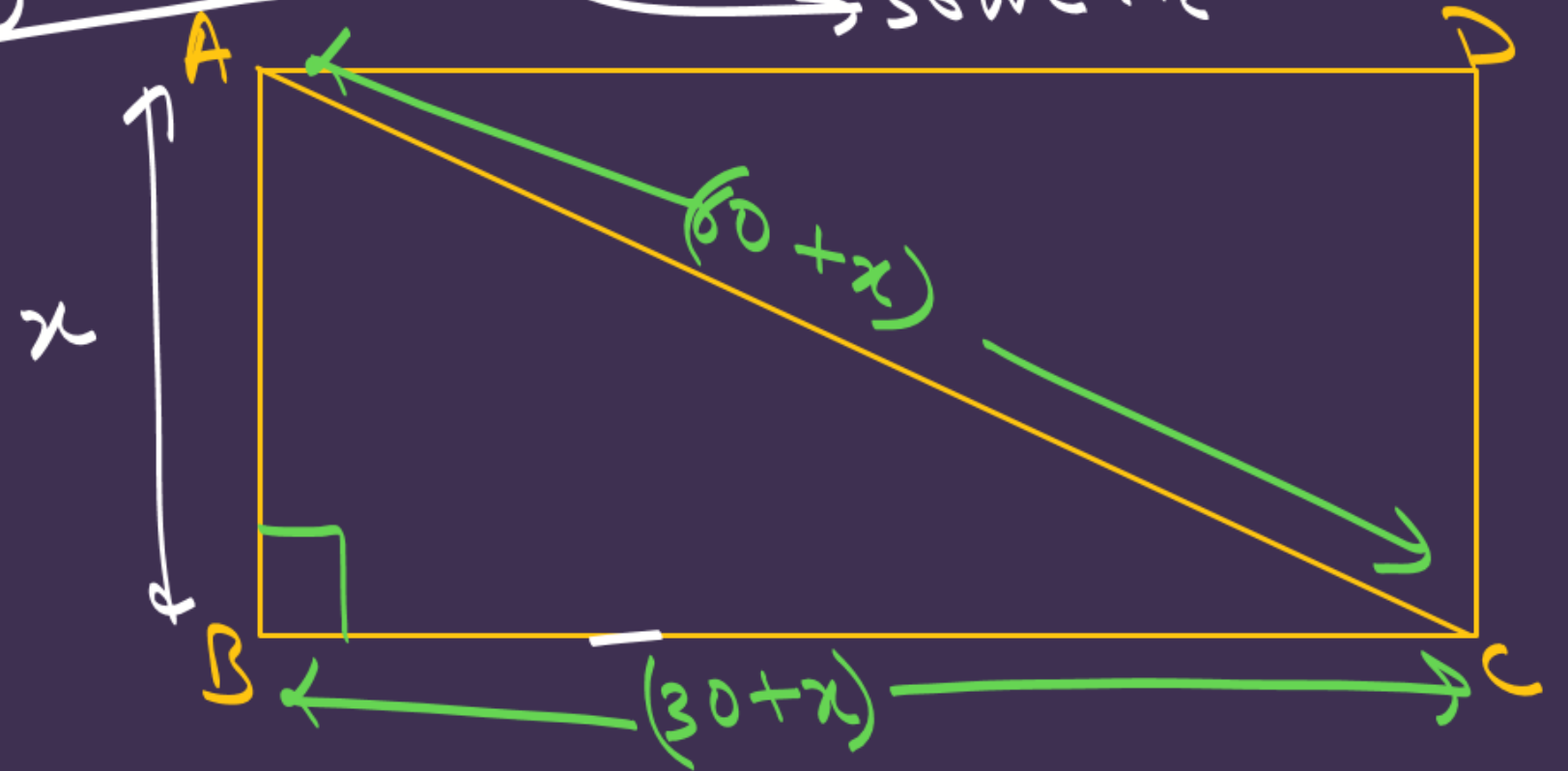
$$AC = (60 + x)$$

ATQ

$$BC = 30 + AB$$

$$BC = (30 + x)$$

~~Pythg~~
 $(60 + x)^2 = x^2 + (30 + x)^2$ → solve ?



#LP: The length of the hall is 5 m more than its breadth. If the area of the floor of the hall is 84 m^2 , what are the length and breadth of the hall?

$$AB = 5 + AD$$

$$\text{let } AD = x \rightarrow B$$

$$\therefore AB = (5+x) \rightarrow L$$

ATQ

$$\text{Area} = 84$$

$$L \times B = 84$$

$$\Rightarrow (x+5)(x) = 84$$

$$\Rightarrow \boxed{x^2 + 5x - 84 = 0}$$

$$\begin{array}{c} +12x \\ -7x \end{array}$$

Solve $\rightarrow x =$



final Ans

$$\begin{array}{l} L = x + 5 = 12 \text{ m} \\ B = x = 7 \text{ m} \end{array} \quad \downarrow$$

#LP: A garden designer is planning a rectangular lawn that is to be surrounded by a uniform walkway.

The total area of the lawn and the walkway is 360 square metres. The width of the walkway is same on all sides. The dimensions of the lawn itself are 12 metres by 10 metres. Based on the information given above, answer the following questions: (CBSE 2025)

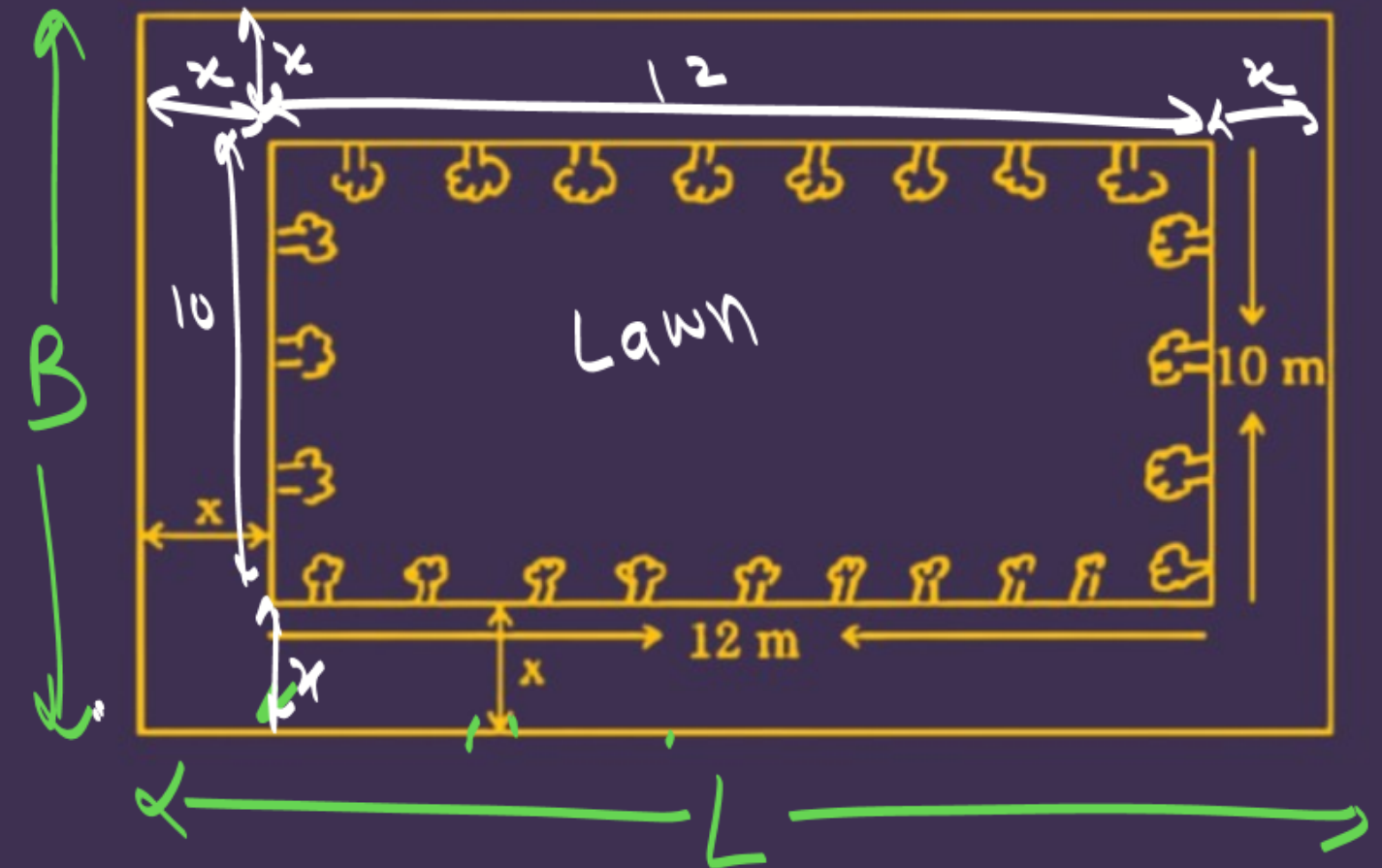
(i) Formulate the quadratic equation representing the total area of the lawn and the walkway, taking width of walkway = x m.

$$\text{Total Area} = 360 \text{ m}^2$$

$L \times B$

$$(12 + 2x)(10 + 2x) = 360$$

Quad ✓



~~(ii)~~ (a) Solve the quadratic equation to find the width of the walkway 'x'.
OR

(b) If the cost of paving the walkway at the rate of ₹ 50 per square metre is ₹ 12,000, calculate the area of the walkway.

(iii) Find the perimeter of the lawn. $\Rightarrow 12 + 10 + 12 + 10$

#LP: A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor. (CBSE 2024)

(i) Assuming the original length of each side of a tile is x units, make a quadratic equation from the above information.



side of sq. tile $\Rightarrow x$ m (let)

Area of floor $\Rightarrow 200 \times (\text{Ar. of 1 tile})$

$$\text{Area of floor} = 200x^2$$



Area of floor $= 128 \times (\text{Ar. of 1 tile})$

$$\text{Area of floor} = 128(x+1)^2$$

same
floor $\Rightarrow 200x^2 = 128(x+1)^2$

↓
quad $\checkmark$

(ii) Write the corresponding quadratic equation in standard form.


$$ax^2 + bx + c = 0$$

(iii) (a) Find the value of x, the length of the side of a tile by factorisation.

OR

(b) Solve the quadratic equation for x using the quadratic formula.



QLP: While designing the school yearbook, a teacher asked the student that the length and width of a particular photo be increased by n units each to double the area of the photo. The original photo is 18 cm long and 12 cm wide. Based on the above information.

Answer the following Questions: (CBSE 2023)

(i) Write an algebraic equation depicting the above information.



(ii) Write the corresponding quadratic equation in standard form.

(iii) What should be the new dimensions of the enlarged photo?

OR

Can any rational value of x make the new area equal to 220 cm^2 ?

HW →

Complete Module

Fri⁶ →

Quad Complete

THANK YOU COODIESS !!

